SAFETY DATA SHEET

M47038 - ANSI - EN

OXY Occidental Chemical Corporation
A subsidiary of Occidental Petroleum Corporation

METHYL CHLORIDE, TECHNICAL GRADE

SDS No.: M47038 **SDS Revision Date:** 19-Dec-2014

1. CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Company Identification:	Occidental Chemical Corporation 5005 LBJ Freeway P.O. Box 809050 Dallas, TX 75380-9050 1-800-752-5151
24 Hour Emergency Telephone Number:	1-800-733-3665 or 1-972-404-3228 (USA); CHEMTREC (within USA and Canada): 1-800-424-9300; CHEMTREC (outside USA and Canada): +1 703-527-3887; CHEMTREC Contract No: CCN16186
To Request an SDS:	MSDS@oxy.com or 1-972-404-3245
Customer Service:	1-800-752-5151 or 1-972-404-3700
Product Identifier:	**METHYL CHLORIDE, TECHNICAL GRADE**
Synonyms:	Chloromethane, Monochloromethane
Product Use:	Silicon manufacturing, Chemical processing
Uses Advised Against:	None identified.

2. HAZARDS IDENTIFICATION

OSHA REGULATORY STATUS: This material is considered hazardous by the OSHA Hazard Communication Standard (29 CFR 1910.1200).

GHS: GERM CELL MUTAGENICITY:	Category 1B - May cause genetic defects
GHS: REPRODUCTION TOXIN:	Category 1B - May damage fertility or the unborn child

GHS SYMBOL:
Flame, Health hazard, Exclamation mark

GHS SIGNAL WORD: **DANGER**

GHS HAZARD STATEMENTS:

GHS - Physical Hazard Statement(s)
Extremely flammable gas

GHS - Health Hazard Statement(s)
May cause drowsiness or dizziness
Harmful if swallowed
Harmful if inhaled
Causes damage to Central Nervous System and Cardiovascular System
Causes damage to Liver, Kidney, and Central Nervous System through prolonged or repeated exposure
May damage fertility or the unborn child
May cause genetic defects

GHS - Precautionary Statement(s) - Prevention
Obtain special instructions before use
Do not handle until all safety precautions have been read and understood
Keep away from heat/sparks/open flames/hot surfaces. — No smoking
Use only outdoors or in a well-ventilated area
Do not breathe gas, fumes, vapor, mist, or spray
Use personal protective equipment as required
Wash thoroughly after handling
Do not eat, drink or smoke when using this product

GHS - Precautionary Statement(s) - Response
Leaking gas fire: Do not extinguish, unless leak can be stopped safely
Eliminate all ignition sources if safe to do so
IF INHALED: Remove person to fresh air and keep comfortable for breathing
Call a POISON CENTER or doctor/physician
IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell
Rinse mouth
Get medical advice/attention if you feel unwell
IF exposed or concerned: Get medical advice/attention
Specific treatment (see Section 4 of the safety data sheet and/or the First Aid information on the product label)

Skin: Direct skin contact may cause frostbite: frozen skin, redness, edema, blisters. Exposure to skin may cause mild skin irritation: redness, dry skin. This material may be absorbed across the skin causing symptoms similar to inhalation exposures.
Eye: Direct eye contact may cause frostbite (opaque cornea, edema). Eye exposure may cause slight eye irritation, pain, conjunctivitis, tearing, clouding of cornea.
Ingestion (Swallowing): Ingestion is not a likely route of exposure because this material is a gas at normal conditions.

Delayed Symptoms/Effects:
- While inhalation is the primary exposure route, the respiratory tract is not generally injured. However, in severe cases inhalation may cause pulmonary congestion
- May cause genetic defects
- May damage fertility or the unborn child

Target Organ Effects: Mild exposures can result in delayed onset of Central Nervous System (CNS) effects (inebriation or drunkenness) that may continue several hours after the exposure ends. Significant and repeat exposures may produce reduced renal output (oliguria), elevation of liver enzymes, renal (kidney) failure, and liver failure. May cause acute renal (kidney) failure and liver failure.

Interaction with Other Chemicals Which Enhance Toxicity: May amplify the effects of other agents that cause Central Nervous System (CNS) and respiratory system depression. Liver toxicity may be enhanced by other agents that cause liver damage, such as alcohol, acetaminophen.

Medical Conditions Aggravated by Exposure: Central Nervous System (CNS) disorders. Liver disorders. Kidney disorders.

Protection of First-Aiders: This is a highly flammable material, handle with extreme care. Avoid contact with the skin and the eyes. Direct contact with liquid or rapidly expanding gas may cause frostbite to contacted tissue (eyes, skin, etc.). Do not breathe gas, fumes, vapor, mist, or spray. Use personal protective equipment. Refer to Section 8 for specific personal protective equipment recommendations. Consider the possibility of high levels of gas in confined/unventilated spaces or low-lying areas.

Notes to Physician: Most cases of intoxication involved concentrations above 500 ppm. Typically, individuals are not uncomfortable during the exposure, and present with nausea, abdominal pain, vomiting, and diarrhea several hours after the exposure. The effects may be delayed and last longer than a similar intoxication with ethanol (alcohol). More common CNS symptoms include drowsiness and fatigue, confusion, headache, ataxia, vertigo, blurred or double vision, tremor, muscular cramping and rigidity, and sleep distrubances. The most severely impacted may enter coma, with or without seizures. Some exposures have been fatal. Epinephrine and other sympathomimetic drugs may initiate cardiac arrhythmias in individuals exposed to this material.

5. FIRE-FIGHTING MEASURES

Fire Hazard: Severe fire hazard. Vapor/air mixtures are explosive. The vapor is heavier than air. Vapors or gases may ignite at distant sources and flash back. Containers may rupture or explode if exposed to heat. Under fire conditions, may produce irritating and/or toxic gases.

Extinguishing Media: Use dry chemical and carbon dioxide.

Methods and Materials for Containment and Cleaning Up:
Take action to protect personnel. Evacuate unnecessary and unprotected personnel. Isolate hazard area and deny entry. ELIMINATE all ignition sources (no smoking, flares, sparks or flames in immediate area). Vapors or gases may ignite at distant ignition sources and flash back. Ventilate closed spaces before entering. Stop leak if possible without personal risk. Shut off ventilation system if needed. Reduce vapors with water spray. Evacuation of surrounding area may be necessary for large spills. Prevent spreading of vapors through sewers, ventilation systems, confined areas. CAUTION: When in contact with refrigerated/cryogenic liquids, many materials become brittle and are likely to break without warning.

Environmental Precautions:
Keep out of water supplies and sewers. Releases should be reported, if required, to appropriate agencies.

7. HANDLING AND STORAGE

Precautions for Safe Handling:
This material is highly flammable, handle with extreme care. Keep away from heat, sparks, flame and other sources of ignition. Ground any equipment used in handling. Use non-sparking tools and equipment. All energized electrical equipment must be designed in accordance with the electrical classification of the area. Do not breathe gas, vapors, or spray mist. Avoid contact with skin, eyes and clothing. Can be absorbed through skin. Direct contact may cause frostbite to exposed tissue (skin, eyes, etc.). Most vapors are heavier than air and will spread along ground and collect in low or confined areas (drains, basements, tanks). Methyl chloride odors are reported not to be noticeable at potentially dangerous concentrations.

Safe Storage Conditions:
Store and handle in accordance with all current regulations and standards. Keep container tightly closed and properly labeled. Store in a cool, dry area. Store in a well-ventilated area. Do not enter confined spaces without following proper confined space entry procedures. Do not enter confined spaces unless adequately ventilated. Avoid heat, flames, sparks and other sources of ignition. Aluminum, zinc or magnesium equipment should not be used for storage or transfer. Grounding and bonding required. May be subject to storage regulations: U.S. OSHA 29 CFR 1910.106. Keep separated from incompatible substances (see below or Section 10 of the Safety Data Sheet).

Incompatibilities/ Materials to Avoid:
Oxidizing Agents (such as Perchlorates, Peroxides, Permanganates, Chlorates, Nitrates, Chlorine, Bromine), Amines, Amides, Magnesium, Sodium, Zinc, Potassium, Aluminum, Alkali metals

GHS: PHYSICAL HAZARDS:
- Flammable Gas - Cat. 1 Extremely Flammable

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Regulatory Exposure Limit(s): Listed below for the product components that have regulatory occupational exposure limits (OEL's).

Component	OSHA Final PEL TWA	OSHA Final PEL STEL	OSHA Final PELCeiling
Methyl Chloride 74-87-3	100 ppm	-----	200 ppm

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state	Compressed, liquefied gas
Color:	Colorless
Odor:	, Ether-like
Odor Threshold [ppm]:	10 ppm. (Methyl chloride odors are reported to be not noticeable at potentially dangerous concentrations.).
Molecular Weight:	50.49
Molecular Formula:	CH3Cl
Decomposition Temperature:	No data available
Boiling Point/Range:	-11.6 °F (-24.2 °C)
Freezing Point/Range:	-144 °F (-97.8 °C).
Melting Point/Range:	Not applicable to liquids
Vapor Pressure:	56.3 psig @ 20 °C (3600 mmHg @ 20 °C)
Vapor Density (air=1):	1.74 @ 0°C
Relative Density/Specific Gravity (water=1):	0.915 @ 20°C
Water Solubility:	0.48 g/100g @ 25 °C
pH:	Not applicable
Volatility:	100%
Evaporation Rate (ether=1):	Not applicable
Partition Coefficient (n-octanol/water):	Log Kow = 0.91
Flash point:	-49.9 °F (-45.5 °C) - OC
Method:	OC - Open Cup
Flammability (solid, gas):	Highly flammable
Lower Flammability Level (air):	8.1 %
Upper Flammability Level (air):	17.4 %
Auto-Ignition Temperature:	1170 °F (632.2 °C)
Viscosity:	No data available

10. STABILITY AND REACTIVITY

Reactivity: Methyl chloride reacts with aluminum to form trimethyl aluminum, a pyrophoric material. Methyl chloride creates an explosive mixture on contact with magnesium, sodium, and other alkali metals. In contact with sodium-potassium alloys, methyl chloride is impact sensitive and may result in an explosion.

Chemical Stability: Methyl chloride is generally stable when dry. In the presence of moisture, methyl chloride hydrolyzes to form corrosive hydrochloric acid. Hydrochloric acid attacks most metals and forms hydrogen gas, which is explosive.

Possibility of Hazardous Reactions:
Avoid heat, flames, sparks and other sources of ignition. Containers may rupture or explode if exposed to heat. Will attack some forms of plastics, rubber, and coatings. Avoid contact with incompatible substances and conditions due to generation of phosgene and other toxic and irritating substances. May react explosively with aluminum in any form. In the presence of moisture, methyl chloride hydrolyzes to form corrosive hydrochloric acid. Methyl chloride creates an explosive mixture on contact with magnesium, sodium, and other alkali metals.

Chronic Effects: Causes damage to the central nervous system, liver, and kidneys through prolonged or repeated exposure. May cause genetic defects. May damage fertility or the unborn child.

SIGNS AND SYMPTOMS OF EXPOSURE:
Listed below.

Inhalation (Breathing): Respiratory System Effects: Central Nervous System (CNS) effects are characteristic following inhalation of chlorinated hydrocarbons and can range from lightheadedness at low level exposures to loss of consciousness at high levels. CNS effects are an early warning that exposure to high levels has occurred and there is risk of cardiac effects (palpitations, low blood pressure, arrhythmia, arrest). CNS effects include the following symptoms: abdominal pain, nausea, vomiting, headache, lightheadedness, blurry or double vision, personality changes, weakness, slurred speech, stupor, incoordination (disequilibrium, ataxia), coma, and respiratory arrest. Onset of symptoms may be delayed from exposure for many hours.
Skin: Direct skin contact may cause frostbite: frozen skin, redness, edema, blisters. Exposure to skin may cause mild skin irritation: redness, dry skin. This material may be absorbed across the skin causing symptoms similar to inhalation exposures.
Eye: Direct eye contact may cause frostbite (opaque cornea, edema). Eye exposure may cause slight eye irritation, pain, conjunctivitis, tearing, clouding of cornea.
Ingestion (Swallowing): Ingestion is not a likely route of exposure because this material is a gas at normal conditions.

TOXICITY:
While the principle exposure route for methyl chloride is inhalation, it can be absorbed through the skin. Methyl chloride is a potent narcotic and has been used as an anesthetic agent. Exposure to less than the current occupational exposure limit (100 ppm, 8 hour TWA) was not associated with any significant impairment. Most cases of intoxication involve concentrations above 500 ppm. In most cases, exposure concentration and duration are not available. The most common consequences of excessive single or repeated exposures have been functional changes in the central nervous system. They have been described as drunkenness as from ingested ethanol (alcohol), but are much longer in persistence. The symptoms of overexposure may include a staggering gait, weakness, drowsiness, double vision, headache, apathy, anorexia, nausea, vomiting, abdominal pain, diarrhea, personality changes, spasms, tremors, loss of memory, paralysis, confusion, unconsciousness, and death. Other organ systems can be affected in persons showing marked central nervous system changes: these include the kidneys, liver, and particularly the lungs. Documentation from historical exposures indicates that pulmonary complications may arise after very significant exposures, likely due to fluid retention from renal failure. Although recovery usually appears complete, at least one case report indicates adverse effects may be permanent. The onset of elevated liver enzymes and indicators of renal impairment may be delayed.

CHRONIC TOXICITY:
Cerebellar damage and neurofunctional impairment and adverse effects on the kidney have been observed in experimental animals. Changes in liver weights and, in some cases, liver damage, have been observed in experimental animals. Studies in experimental animals have shown adverse effects on the spleen in several species. Effects on the spleen have usually been at neurotoxic levels of exposure; therefore, the relevance to humans is not clear.

Interaction with Other Chemicals Which Enhance Toxicity: May amplify the effects of other agents that cause Central Nervous System (CNS) and respiratory system depression. Liver toxicity may be enhanced by other agents that cause liver damage, such as alcohol, acetaminophen.

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GHS HEALTH HAZARDS:

BIOCONCENTRATION: An estimated BCF of 3 suggests the potential for bioconcentration in aquatic organisms is low.

13. DISPOSAL CONSIDERATIONS

Waste from material:
Use or reuse if possible. Dispose in accordance with all applicable regulations. May be subject to disposal regulations.

Container Management:
Dispose of container in accordance with applicable local, regional, national, and/or international regulations. Container rinsate must be disposed of in compliance with applicable regulations.

14. TRANSPORT INFORMATION

LAND TRANSPORT

U.S. DOT 49 CFR 172.101:

UN NUMBER:	UN1063
PROPER SHIPPING NAME:	Methyl chloride
HAZARD CLASS/ DIVISION:	2.1
LABELING REQUIREMENTS:	2.1
RQ (lbs):	RQ 100 Lbs. (Methyl chloride)

CANADIAN TRANSPORTATION OF DANGEROUS GOODS:

UN NUMBER:	UN1063
SHIPPING NAME:	Methyl chloride
CLASS OR DIVISION:	2.1
LABELING REQUIREMENTS:	2.1

MARITIME TRANSPORT (IMO / IMDG) Regulated

UN NUMBER:	UN1063
PROPER SHIPPING NAME:	Methyl Chloride
HAZARD CLASS / DIVISION:	2.1
LABELING REQUIREMENTS:	2.1

Reason for Revision:
- Three year review
- Changed the SDS format to meet the GHS requirements of the revised 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)
- Updated the (M)SDS header
- Updated 24 Hour Emergency Telephone Number: SEE SECTION 1
- Product Identifier has been added or updated: SEE SECTION 1
- Updated Uses Advised Against information: SEE SECTION 1
- Added OSHA Status: SEE SECTION 2
- Emergency Overview was revised: SEE SECTION 2
- Added GHS Information: SEE SECTION 2
- Added synonym(s): SEE SECTION 3
- Updated First Aid Measures: SEE SECTION 4
- Revised Accidental Release Measures: SEE SECTION 6
- Revised Handling and Storage Recommendations: SEE SECTION 7
- PPE recommendations have been modified: SEE SECTION 8
- Updated Physical and Chemical Properties. SEE SECTION 9
- Stability and Reactivity recommendations: SEE SECTION 10
- Toxicological information has been revised: SEE SECTION 11
- Ecological Information has been modified: SEE SECTION 12
- Updated Disposal Considerations. SEE SECTION 13
- Updated Transportation Information: SEE SECTION 14
- Regulatory Information Changes: SEE SECTION 15
- Revised Preparer Information: SEE SECTION 16
- Added SDS Revision Date: SEE SECTION 16
- Added/Updated Revision Log: SEE SECTION 16
- Added "End of Safety Data Sheet" phrase

IMPORTANT:
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OSHA Standard 29 CFR 1910.1200 requires that information be provided to employees regarding the hazards of chemicals by means of a hazard communication program including labeling, safety data sheets, training and access to written records. We request that you, and it is your legal duty to, make all information in this Safety Data Sheet available to your employees

End of Safety Data Sheet